Chapter 12 introduces Amazon DynamoDB, a NoSQL database offered by AWS. DynamoDB is typically not compatible with legacy applications. You need to rework your applications to use DynamoDB. You'll implement a to-do application in this chapter.

Chapter 13 explains what's needed to make your infrastructure highly available. You'll learn how to recover automatically from a failed virtual machine or even a whole data center.

Chapter 14 introduces the concept of decoupling your system to increase reliability. You'll learn how to use synchronous decoupling with the help of Elastic Load Balancing (ELB). Asynchronous decoupling is also part of this chapter; we explain how to use the Amazon Simple Queue Service (SQS), a distributed queuing service, to build a fault-tolerant system.

Chapter 15 introduces three different ways to deploy software to AWS. You'll use each of the tools to deploy an application to AWS in an automated fashion.

Chapter 16 dives into building fault-tolerant applications based on the concepts explained in chapters 13 and 14. You'll create a fault-tolerant image-processing web service within this chapter.

Chapter 17 is all about flexibility. You'll learn how to scale the capacity of your infrastructure based on a schedule or based on the current load of your system.

Chapter 18 explains ways to deploy containers on AWS. You'll learn to use ECS with Fargate and App Runner to run your containerized application.